Vishanth Hari Raj Balasubramanian

US | ☎ (310)-254-6562 | vishanth.hari.raj.ucla@gmail.com | 🔗 Website | 🔗 LinkedIn | 🔗 GitHub

SUMMARY

Staff Machine Learning Engineer with ***5+ years' of SDE, MLOps & Applied Machine Learning*** experience at ***F500 & Startup***, leading **AI Guardrails / LLM Security R&D** at Zscaler, proven track record of innovation impacting ***$5M+*** in business value, expertise in **LLM inference optimization**, GenAI, AI Agents, RAG, multimodal, NLP, LLMs, MLOps, ETL data pipelines, Agile (SDLC), CI/CD. Proficient in *SQL, Python, C++, AWS, Azure*. *STEM Graduate with **M.S** @UCLA, **B.S** @NITT*

SKILLS

Programming:	Python (PyTorch, TensorFlow, MLflow, NLTK), C++, JavaScript, SQL, NoSQL.
Inference:	TensorRT, TensorRT-LLM, ONNX Runtime, vLLM, llama.cpp, CUDA, Tensor/Pipeline Parallelism, FSDP, Quantization (FP8)
Tools:	Git, Jira, Airflow, Hadoop, Spark, Snowflake, AWS, LangChain, LangGraph, Kubernetes, Docker, Streamlit
Certification:	***Lean Six Sigma Green Belt***, Machine Learning using AWS, Apache Spark, DevOps Leadership
Model:	**LLM**, AI Guardrails, LoRA Fine-tuning, XGBoost, RAG, NLP, CNN, RNN, GAN, Transformers, Random Forest, A/B Test.

EXPERIENCE

ZSCALER — San Jose, USA
Staff Machine Learning Engineer – AI Guard — Oct 2026 – Present
Senior Machine Learning Engineer – AI Guard — Dec 2025 – Oct 2026

- **Leading AI Guard ML R&D:** built & **fine-tuned AI guardrail models** (encoder & decoder LLMs, **LoRA**) to detect ***jailbreak & prompt-injection*** attacks and LLM data leakage for enterprise AI applications
- **Optimized GPU inference** for production guardrail models using **TensorRT, ONNX & vLLM**, achieving **2× RPS** and **~50% lower latency**, **saving ~$2M per year** in serving cost
- **Benchmarked GPUs** (NVIDIA, AMD) for self-hosted models; **built MLOps** taking AI Guard detectors to production

CORNAMI INC — San Jose, USA
AI/ML Software Engineer – GenAI — Feb 2025 – Dec 2025

- Developed & optimized ***LLM inference engines***, leveraging flash attention, *KV cache & quantization FP8* to reduce latency & improve TPS by **30%** for inference on **LLM Models** (Llama 3.3 70B, Llama 4, DeepSeek)
- **Implemented distributed computing techniques** with **Tensor Parallelism, Pipeline Parallelism**, scaling inference across multiple GPUs (DGX H100 servers) & proprietary fractal cores, replicating low-level algorithms through CUDA
- **Optimized & validated LLM inference** on H100 GPUs and Fractal cores, increasing **60% TPS** and reducing **10% latency** using continuous batching
- **Benchmarked** LLMs on TensorRT-LLM, Ollama & vLLM (prefill, memory, latency, TPS) to drive optimization strategy

TRAVERSAAL AI — Los Angeles, USA
Machine Learning Engineer Intern — June 2024 – Sep 2024

- Developed **Enterprise LLM Search Engine** Chatbot E2E application integrated with **RAG (Retrieval-Augmented Generation)** multimodal model format, *used LLM models (Llama 3.1, BART, nomic-v3)* for clients, *enhanced information retrieval by **30%.***
- **Trained/Fine-tuned** pre-trained LLM using **LoRA** with domain knowledge & designed a ***knowledge graph*** with semantic chunking, improved data retrieval efficiency by 20%, reduced search time by **500s**
- **Stack:** *LangChain, Hugging Face, Llama, BERT, AWS SageMaker, SQL, APIs, FAISS, LangGraph, OpenAI API, Vertex AI*

CITIGROUP — Chennai, India
Software Engineer 2 – AI/ML — July 2021 – Sep 2023

- **Innovation Lead** for HNI AI/ML: spearheaded ML models *to forecast incidents (in Citi Payments) and analyse downtime impact*, *reduced incidents* impact by **15%, saved $700K+**
- **Integrated LLM/NLP *models*** *into customer chatbots* for automatic incident creation, LoB assignment & sentiment analysis; **improved resolution time by 30%**
- Developed **ML Models** using ***XGBoost, Random Forest*** to predict *app login & payment failures* from real-time logs & **Citi 3DS** data, prevented **50+** major incidents, boosting revenue by **~$1.1M**
- **Implemented recommendation system** (**Random Forest**, **TF-IDF**) suggesting incident-resolution actions from past incidents, improving resolution time by **10%**
- **Architected** robust **MLOps** pipeline & infrastructure; **deployed APIs** integrating ML endpoints with dashboards, **reducing latency by 5 mins** on SNOW & Splunk data streams
- **Stack:** *ML models (TensorFlow, Scikit-learn), NLP, LLM, Kubernetes, ReactJS, REST APIs, SQL, microservices, Linux*

EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA) — Los Angeles, USA
*Master of Science, Business Analytics – Data Science/Machine Learning **(STEM)*** — Dec 2024

NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPALLI (NIT-T) — Trichy, India
*Bachelor of Technology, Computer Science/Industrial Engineering **(Institute Rank Holder)*** — June 2021

RESEARCH/PROJECTS

Multimodal Hotel Search Engine Chatbot : ***fine-tuned BART*** for review summarization; embed pipeline reduced search time by 20%
GenAI Agents Auto-Interviews: Built **AI Agent** (crewai) using ***whisper-large-v3-turbo*** for an audio interview chatbot with ratings
Detect fake images generated by AI: Built **multi-modal GAN** combining image/text features (ResNet, BERT), **accuracy: 97.6%**
Detection of Cars/Traffic Signal : Ensemble model (*ResNet, ConvNext*) with additional dense layers achieving ***MSE of 0.02***
Weather Prediction, LSTM/GRU: Built a ***bidirectional LSTM model*** predicting weather trends across cities, achieving an ***MSE of 0.012***